Mini Project 2

Complete C++ Program

```
#include <iostream>
#include <fstream>
#include <vector>
#include <string>
using namespace std;

class Content {
public:
  string title, platform, status;
  int views;

  void display() {
    cout << "Title: " << title << endl;
    cout << "Platform: " << platform << endl;
    cout << "Views: " << views << endl;
    cout << "Status: " << status << endl;
  }
};

// Read all content from file
vector<Content> readContent() {
  vector<Content> list;
  ifstream file("content_list.txt");
  Content c;
```

```
    while (getline(file, c.title) &&
           getline(file, c.platform) &&
           file >> c.views &&
           file.ignore() &&
           getline(file, c.status)) {
        list.push_back(c);
    }

    file.close();
    return list;
}

// Display content in numbered list
void displayList() {
    vector<Content> list = readContent();

    if (list.empty()) {
        cout << "No content found.\n";
        return;
    }

    for (int i = 0; i < list.size(); i++) {
        cout << i + 1 << ". "
             << list[i].title << " - "
             << list[i].platform << endl;
    }
}
```

```
// Add new content
void addContent() {
    Content c;

    cin.ignore();
    cout << "Enter title: ";
    getline(cin, c.title);

    cout << "Enter platform: ";
    getline(cin, c.platform);

    cout << "Enter views: ";
    cin >> c.views;
    cin.ignore();

    cout << "Enter status: ";
    getline(cin, c.status);

    ofstream file("content_list.txt", ios::app);

    file << c.title << endl;
    file << c.platform << endl;
    file << c.views << endl;
    file << c.status << endl;

    file.close();
```

```
    cout << "Content saved successfully!\n";
}

// Update content status
void updateStatus() {
    vector<Content> list = readContent();
    displayList();

    if (list.empty()) return;

    int number;
    cout << "Enter content number: ";
    cin >> number;
    cin.ignore();

    if (number < 1 || number > list.size()) {
        cout << "Invalid number!\n";
        return;
    }

    cout << "Enter new status: ";
    getline(cin, list[number - 1].status);

    ofstream file("content_list.txt");

    for (Content c : list) {
        file << c.title << endl;
        file << c.platform << endl;
```

```
        file << c.views << endl;
        file << c.status << endl;
    }

    file.close();
    cout << "Status updated successfully!\n";
}

// Delete content
void deleteContent() {
    vector<Content> list = readContent();
    displayList();

    if (list.empty()) return;

    int number;
    cout << "Enter content number to delete: ";
    cin >> number;

    if (number < 1 || number > list.size()) {
        cout << "Invalid number!\n";
        return;
    }

    list.erase(list.begin() + number - 1);

    ofstream file("content_list.txt");
```

```
    for (Content c : list) {
        file << c.title << endl;
        file << c.platform << endl;
        file << c.views << endl;
        file << c.status << endl;
    }

    file.close();

    cout << "Content deleted successfully!\n";
    cout << "\nUpdated Content List:\n";
    displayList();
}

int main() {
    int choice;

    do {
        cout << "\n===== CONTENT MANAGEMENT SYSTEM =====\n";
        cout << "1. Add Content\n";
        cout << "2. Display Content List\n";
        cout << "3. Update Content Status\n";
        cout << "4. Delete Content\n";
        cout << "5. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
```

```
        case 1:
            addContent();
            break;

        case 2:
            displayList();
            break;

        case 3:
            updateStatus();
            break;

        case 4:
            deleteContent();
            break;

        case 5:
            cout << "Program ended.\n";
            break;

        default:
            cout << "Invalid choice!\n";
    }

  } while (choice != 5);

  return 0;
}
```

Explanation of the 5 tasks

1. Content class: Stores the title, platform, views, and status. The display() method prints all details.
2. Add content: Takes input and uses ofstream with ios::app to add data without deleting existing entries.
3. Display content: Uses ifstream to read the file and show each item with its number, title, and platform.
4. Update status: Reads all items into a vector, changes the selected status, and rewrites the file.
5. Delete content: Removes the selected item using erase() and saves the remaining items back to the file.

Example output

===== CONTENT MANAGEMENT SYSTEM =====

1. Add Content

2. Display Content List

3. Update Content Status

4. Delete Content

5. Exit

Enter your choice: 1

Enter title: My First Video

Enter platform: YouTube

Enter views: 1500

Enter status: Planned

Content saved successfully!